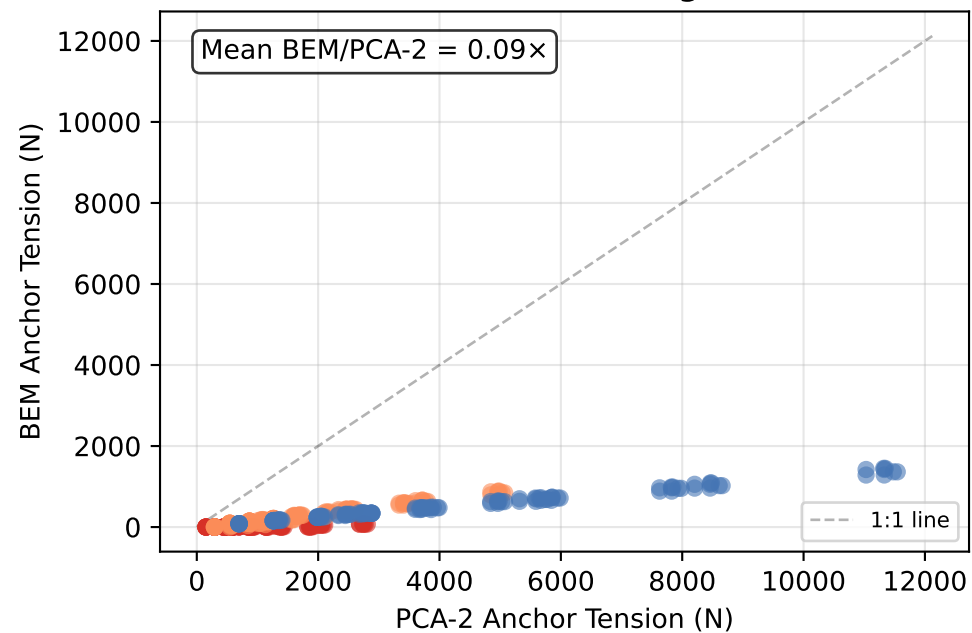


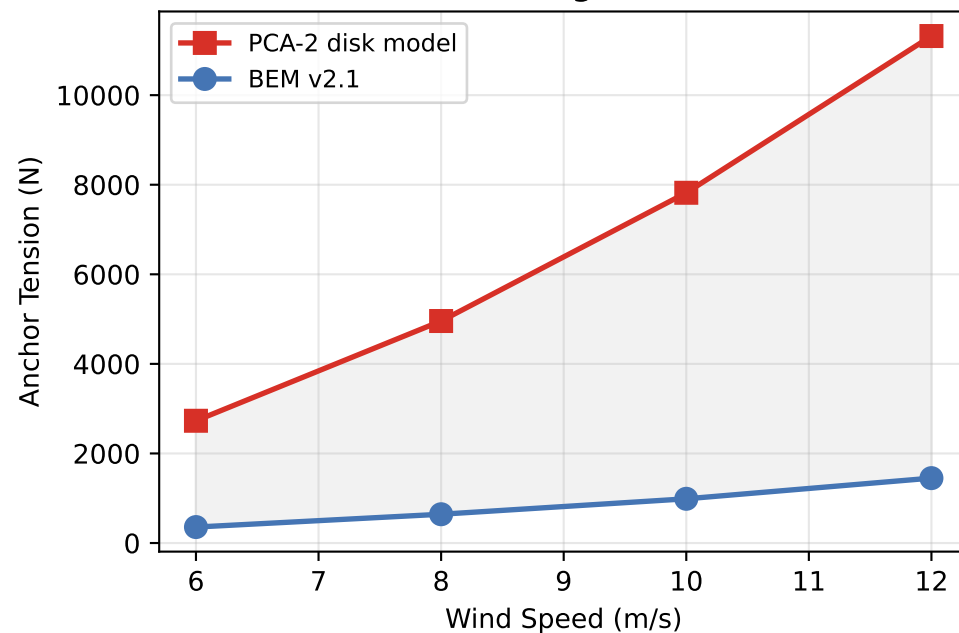
PCA-2 Disk Model vs BEM v2.1: What Did We Gain?

Identical Configurations, Different Aerodynamic Models

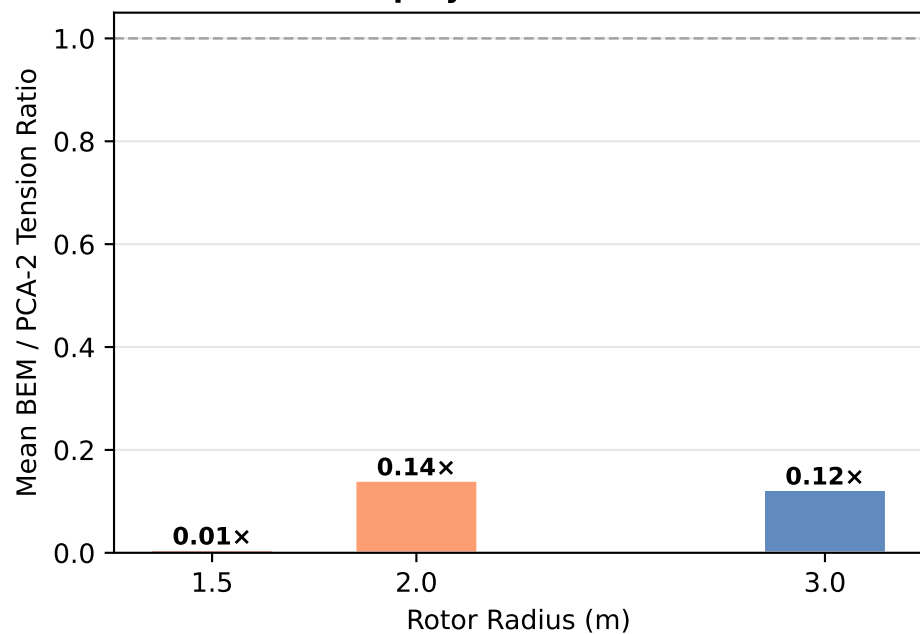
BEM vs PCA-2: All Configurations



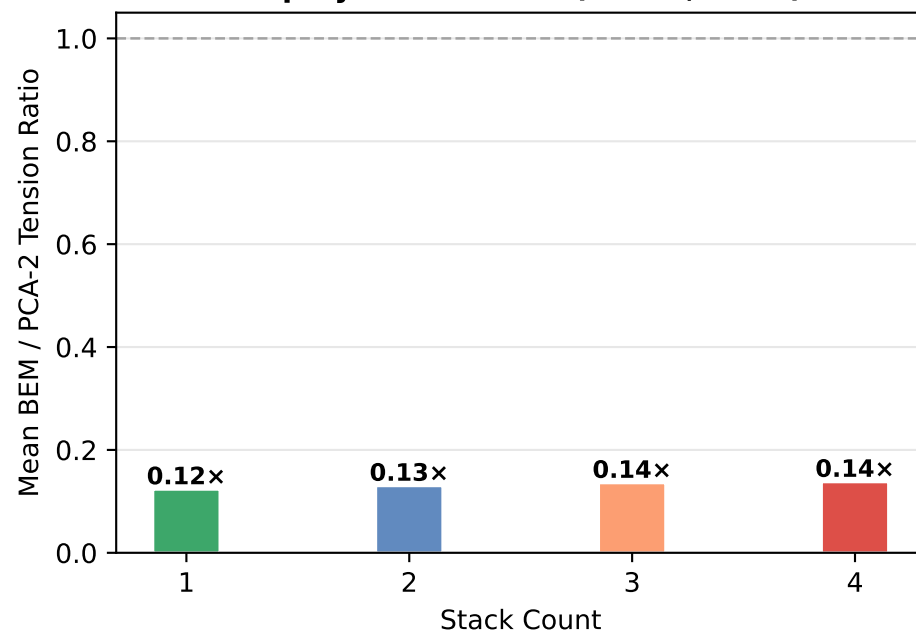
R=3m, N=4, graded, 45°



Gap by Rotor Radius



Gap by Stack Count (R=2.0, 3.0m)



IMPLICATIONS: The PCA-2 disk model systematically overestimates forces by ~11x compared to BEM v2.1. The gap is largest at R=1.5m (~163x) and narrows at R=3.0m. This is the solidity + 2-D/3-D averaging gap (see DECISIONS.md D1). The PCA-2 empirical data represents a high-solidity disk ($\sigma \approx 0.098$) while BEM models our low-solidity blades ($\sigma \approx 0.032$). BEM is the design truth for our configuration. Data: identical sweep grid, 45° and 55° elevation.